

Coursework 2 Brief (Group)

Overview

This coursework series is designed to develop your practical skills in machine learning through structured, comparative experiments on a selected benchmark dataset using **scikit-learn**. You will demonstrate your understandings of dataset investigation, analysis design, performance evaluation, and overfitting prevention while focussing on **experiment design and justification rather than raw accuracy**.

This coursework series prioritises:

- Rigorous **experiment design**
- Clear **justification of modelling choices**
- Meaningful **performance analysis** over raw accuracy

Task

Apply **2-3 scikit-learn tools** that best suit the benchmark task, critically compare the tools, and suggest and justify the best-performing pipeline.

Weighting

1. **20%** Group report
2. **10%** Oral presentation

Submission

All submissions will be made in group.

1. The **report** (maximum 15 pages) should include:
 - Design of experiments and pipelines
 - Comparative analysis of accuracy, computational efficiency, and overfitting handling
 - Justification for the suggested pipeline

NB The report must use the **provided template**.

2. The **presentation** will deliver 10-minute oral presentation and 10-minute Q&A.
 - Succinctly communicate comparative analysis findings (accuracy, computational efficiency, overfitting handling).
 - Justify and clearly explain your chosen best pipeline.
 - Answer questions to demonstrate understanding.
 - To demonstrate teamwork, you do not need to take turns speaking, but your presentation must clearly explain the distinctive contributions made by each member.

Submission Format

- All reports must be submitted in PDF, using the **provided template**.
- Each report must be accompanied by a **Jupyter Notebook**, showing your workflow and outputs clearly.
- Ensure your notebooks are clean and reproducible.
- Group submissions need to be titled “ENG4200_CW<#>_report_Group<group ID>.pdf” and “ENG4200_CW<#>_presentation_<group ID>.pdf”.
 - Ensure that in the group submissions, include all group members’ names and student IDs in **the first page** (both for report and presentation).

Assessment Criteria (Applies to All Coursework)

Report

Grade Band	1. Design of experiments and pipelines (25%)	2. Comparative Analysis of Accuracy and Overfitting Handling (25%)	3. Comparative Analysis of Computational Efficiency (25%)	4. Justification of Suggested Pipeline (25%)
A Excellent	Comprehensive and well-structured experimental design. Pipelines are thoughtfully constructed, appropriate for the task, and make full use of relevant preprocessing and model tuning strategies. Evidence of creativity and rigour in approach.	Thorough, well-justified comparison of all models using relevant accuracy metrics. Overfitting is actively analysed and mitigated using appropriate techniques (e.g. cross-validation), with strong empirical support.	Clear and critical analysis of computational efficiency, including training time, scalability, and resource usage across models. Strong empirical evidence supports conclusions.	Final pipeline is well-chosen, clearly justified, and supported by empirical comparison. Trade-offs between models are critically evaluated, with context-sensitive reasoning.
B Very Good	Sound design of experiments and pipelines. Choices are mostly appropriate and well-explained, though some steps may lack optimisation or clarity. Covers all required components.	Comparison is clear and uses suitable metrics. Some attempt to analyse overfitting, with use of appropriate strategies. Evidence is generally convincing but may lack depth in places.	Good analysis of computational aspects. Covers training time and basic scalability issues. Conclusions are reasonable, though may not be fully explored.	Final pipeline is appropriate and justified. Good comparison of alternatives with some critical insight. Minor gaps in rationale or supporting evidence.
C Good	Adequate pipeline setup. Basic design and experimentation are present, though some parts may be inconsistent, incomplete, or poorly justified.	Basic comparison of models using accuracy metrics. Some mention of overfitting, but analysis is limited or lacks justification. Mitigation strategies may be generic or underdeveloped.	Basic analysis of runtime or resource use. Limited or mostly descriptive comparison. Some claims lack supporting evidence.	Pipeline choice is reasonable but lacks strong justification. Trade-offs are mentioned but not critically analysed.
D Satisfactory	Minimal or inconsistent pipeline design. Limited experimentation. Key stages (e.g. preprocessing or validation) are missing or flawed.	Superficial comparison. Metrics used without clear explanation or consistency. Overfitting not meaningfully addressed.	Minimal or flawed discussion of efficiency. Some reference to speed or complexity, but lacks clarity or analysis.	Pipeline is selected with little justification. Comparison is shallow or mostly descriptive.
E Weak	Pipeline setup is missing or fundamentally flawed. No evidence of meaningful experimentation or understanding of pipeline structure.	No meaningful comparison of accuracy. Overfitting issues not identified or misunderstood.	No relevant analysis of computational efficiency.	Pipeline is inappropriate or unjustified. Choice appears arbitrary or unsupported.

Presentation

Grade Band	1. Clarity of Communication (25%)	2. Depth and Accuracy of Content (25%)	3. Justification of Suggested Pipeline (25%)	5. Engagement and Q&A (25%)
A (18–22) Excellent	Clear, confident, well-paced delivery using accessible language and logical flow.	Accurate, insightful comparative analysis with clear evidence and correct terminology.	Justification is clear, well-reasoned, and demonstrates critical evaluation.	Confident, accurate responses demonstrating deep understanding.
B (15–17) Very Good	Mostly clear and confident with minor lapses in flow or clarity.	Very good analysis with strong accuracy and minor omissions.	Clear justification with good reasoning.	Clear, mostly accurate responses demonstrating good understanding.
C (13–14) Good	Generally clear but some unclear segments or pacing issues.	Good content with some gaps or inaccuracies.	Justification present but basic.	Responses mostly clear with minor inaccuracies.
D (11–12) Satisfactory	Understandable but lacks clarity or confident delivery.	Satisfactory but lacks depth; inaccuracies present.	Limited or unclear justification.	Responses basic, with some confusion or incorrect answers.
E (7–10) Weak	Unclear delivery; difficult to follow.	Weak or inaccurate content with significant omissions.	No clear justification provided.	Unable to answer questions appropriately.